

ASSIUT INTL, Egypt

WMO: 623930

Lat: 27.047N

Lon: 31.012E

Elev: 235

StdP: 98.53

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	5.9	-5.9	2.4	16.2	-3.9	2.8	16.5	10.3	16.8	9.4	16.1	2.2	330	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	41.7	21.1	40.1	21.0	38.8	20.8	23.8	36.1	23.0	35.4	22.3	34.7	5.1	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.3	28.9	19.1	14.3	28.6	18.1	13.4	28.6	72.4	36.0	69.2	35.1	66.5	34.6	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.1	8.3	DB	2.5	44.9	1.4	1.9	1.5	46.3	0.6	47.3	-0.1	48.4	-1.2	49.7	(4)
(5)			WB	0.0	25.1	1.4	1.8	-1.0	26.4	-1.8	27.4	-2.6	28.4	-3.6	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	13.0	15.1	18.9	23.3	27.6	29.9	30.9	30.8	28.4	24.7	19.3	14.5	(6)
(7)		DBStd	7.01	2.43	3.50	3.96	3.77	3.35	2.48	1.85	1.72	2.16	2.87	2.93	2.50	(7)
(8)		HDD10.0	4	3	1	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	457	168	104	39	2	0	0	0	0	0	0	22	122	(9)
(10)		CDD10.0	4774	94	143	276	399	546	598	647	644	552	454	278	141	(10)
(11)		CDD18.3	2186	1	14	57	152	287	348	388	386	302	197	49	4	(11)
(12)		CDH23.3	29922	40	210	786	1902	3934	5106	5846	5713	3829	1999	483	75	(12)
(13)		CDH26.7	17274	5	76	363	981	2317	3161	3644	3509	2117	921	160	20	(13)
(14)	Wind	WSAvg	4.1	3.5	3.9	4.4	4.4	4.2	4.8	4.2	4.0	4.5	4.1	3.5	3.6	(14)
(15)	Precipitation	PrecAvg	8	1	0	2	0	0	0	0	0	0	4	0	(15)	
(16)		PrecMax	60	8	1	11	1	1	0	0	0	1	52	9	(16)	
(17)		PrecMin	1	0	0	0	0	0	0	0	0	0	0	0	(17)	
(18)		PrecStd	11	1	0	3	0	0	0	0	0	0	10	2	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	32.5	37.7	40.5	43.8	43.9	42.3	42.1	40.8	38.6	34.1	29.1	(19)
(20)			MCWB	14.1	16.2	17.7	18.7	21.0	21.6	21.7	22.7	21.5	20.0	18.0	16.0	(20)
(21)		2%	DB	24.1	29.2	33.9	37.9	40.8	41.2	40.3	40.0	38.6	35.8	30.8	25.1	(21)
(22)			MCWB	13.2	14.8	16.5	18.2	20.7	20.8	21.5	21.8	21.2	20.1	17.6	14.6	(22)
(23)		5%	DB	21.6	25.7	30.9	35.1	38.2	39.2	38.9	38.6	36.8	33.7	28.1	22.2	(23)
(24)			MCWB	12.3	13.7	15.7	17.2	19.7	20.6	21.2	21.5	21.2	19.8	16.9	13.6	(24)
(25)		10%	DB	19.2	22.9	27.6	32.4	36.1	37.4	37.6	37.1	35.0	31.5	26.0	20.5	(25)
(26)			MCWB	11.5	12.8	14.6	16.4	18.8	20.3	20.9	21.1	20.7	19.0	16.2	13.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	17.5	18.9	20.2	23.2	24.1	25.8	24.6	24.0	22.6	20.0	17.4	(27)
(28)			MCD B	24.2	30.3	33.3	36.5	39.2	38.4	34.9	36.5	36.1	33.3	29.5	26.6	(28)
(29)		2%	WB	13.8	15.6	17.2	18.9	21.8	22.7	23.7	23.6	22.9	21.3	18.6	15.4	(29)
(30)			MCD B	22.1	26.4	31.3	34.7	37.4	37.2	35.2	35.6	34.3	32.2	27.2	22.9	(30)
(31)		5%	WB	12.7	14.3	16.1	17.9	20.4	21.6	22.7	22.8	22.1	20.4	17.6	14.2	(31)
(32)			MCD B	20.1	24.1	29.5	33.1	35.5	36.1	34.8	34.9	33.1	31.1	26.4	21.2	(32)
(33)		10%	WB	11.8	13.0	14.9	16.9	19.5	20.8	21.9	22.1	21.3	19.6	16.6	13.3	(33)
(34)			MCD B	18.9	22.0	26.6	31.1	34.4	34.8	34.2	34.3	32.3	29.8	25.1	19.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	13.8	14.6	15.0	15.0	14.3	13.4	13.2	13.2	13.3	12.9	(35)	
(36)			MCDBR	16.4	17.8	18.9	18.6	17.9	16.7	14.9	14.7	15.3	15.4	15.8	15.2	(36)
(37)			MCWBR	8.4	8.4	8.1	7.3	7.1	5.8	5.4	5.3	5.4	5.5	6.9	7.7	(37)
(38)		5% WB	MCD BR	14.7	16.3	17.4	17.1	16.4	15.0	13.6	13.6	13.5	13.8	14.3	13.8	(38)
(39)			MCWBR	7.9	8.1	7.8	7.2	6.8	5.9	5.6	5.4	5.5	5.5	6.7	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.384	0.416	0.460	0.542	0.558	0.483	0.473	0.464	0.467	0.469	0.401	0.384	(40)	
(41)		taud	2.085	1.982	1.864	1.697	1.687	1.882	1.929	1.958	1.935	1.912	2.065	2.094	(41)	
(42)		Ebn at Noon	844	847	831	775	761	816	823	829	814	782	817	823	(42)	
(43)		Edh at Noon	140	168	200	243	246	201	192	185	184	178	142	133	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.34	5.28	6.50	7.36	7.87	8.30	8.07	7.56	6.78	5.62	4.61	4.04	(44)	
(45)		RadStd	0.11	0.19	0.20	0.16	0.18	0.08	0.08	0.09	0.14	0.09	0.10	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+2.65	N/A	+1.29	+1.14	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.40	+0.36	N/A	N/A	N/A	+0.46	N/A	N/A	N/A	N/A

Nomenclature: See separate page